Share Issuance Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Share Issuance Restraint (SIR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on SIR achieves an annualized gross (net) Sharpe ratio of 0.51 (0.45), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 28 (25) bps/month with a t-statistic of 3.60 (3.27), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (5 year), Growth in book equity, Share issuance (1 year), Momentum and LT Reversal, Long-term EPS forecast, Analyst Value) is 20 bps/month with a t-statistic of 2.32.

1 Introduction

Production-based asset pricing models fundamentally link cross-sectional variation in expected returns to firms' production decisions and the frictions they face in capital markets. While traditional models emphasize how firms' investment policies and financing constraints shape their exposure to systematic risks, the precise mechanisms through which corporate financing decisions—particularly equity issuance restraint—relate to risk premia remain incompletely understood. Firms that refrain from issuing equity despite growth opportunities may signal either operational efficiency that reduces their need for external capital or binding financial frictions that constrain their ability to access equity markets. This distinction carries profound implications for understanding how production technologies and adjustment costs translate into cross-sectional return predictability.

In a production economy, firms' equity issuance decisions directly reflect the interaction between marginal productivity of capital and the shadow cost of external financing (Cochrane, 1991; Zhang, 2005). When firms exhibit share issuance restraint (SIR), they reveal information about their production function characteristics and their exposure to aggregate productivity shocks. Firms with high SIR operate with lower marginal adjustment costs, as their ability to fund investment internally suggests more efficient conversion of retained earnings into productive capital (Li and Zhang, 2010). This operational efficiency reduces these firms' sensitivity to financial market frictions and aggregate credit supply shocks, leading to lower conditional risk premia in equilibrium.

The production-based framework predicts that SIR proxies for firms' exposure to investment-specific technology shocks that drive time-variation in the marginal efficiency of investment (Papanikolaou, 2011). High-SIR firms, by avoiding dilutive equity issuance, demonstrate resilience to fluctuations in the cost of capital adjustment and maintain more stable production paths during periods of technological disruption. These firms effectively self-insure against productivity shocks through their financing policies, reducing their covariance with the stochastic discount factor in a production economy (Berk and Green, 2004).

Furthermore, share issuance restraint maps directly to cross-sectional differences in firms' investment-capital ratios and their implied returns under q-theory (Liu et al., 2009; Hou et al., 2015). Firms exhibiting high SIR maintain lower investment rates relative to their capital stock, consistent with operating in the region of their production function where marginal q approximates average q. This production equilibrium implies that high-SIR firms face lower costs of capital in competitive markets, as their restraint signals proximity to optimal scale rather than growth constraints. The resulting cross-sectional spread in expected returns between high- and low-SIR firms emerges endogenously from heterogeneity in production technologies and the convexity of adjustment cost functions.

We find that a value-weighted long/short trading strategy based on Share Issuance Restraint achieves an annualized gross Sharpe ratio of 0.51 and generates a monthly average excess return of 30 basis points with a t-statistic of 4.01. The strategy's performance remains robust across multiple factor model specifications, with the monthly alpha relative to the Fama-French six-factor model equaling 28 basis points (t-statistic = 3.60). These results demonstrate significant economic magnitude, as the SIR strategy's gross Sharpe ratio exceeds 90% of documented anomalies in the factor zoo, positioning it among the most reliable cross-sectional return predictors.

The strategy exhibits remarkable consistency across different portfolio construction methodologies and maintains statistical significance after accounting for transaction costs. Net of trading costs, the value-weighted quintile strategy delivers a monthly average return of 26 basis points (t-statistic = 3.49) and achieves positive generalized alphas relative to all five standard factor models tested. Notably, the strategy's performance persists among large-cap stocks, where the long/short SIR portfolio within the largest size quintile generates an average monthly return of 21 basis points (t-statistic = 2.36), alleviating concerns about small-stock illiquidity driving the results.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the SIR trading strategy maintains a statistically significant alpha of 20 basis points per month (t-statistic = 2.32). The strategy loads positively on the CMA factor (coefficient = 0.18, t-statistic = 3.28), consistent with the production-based interpretation that high-SIR firms operate with conservative investment policies that reduce their exposure to investment-specific shocks. These spanning test results confirm that Share Issuance Restraint captures a distinct dimension of cross-sectional return variation not subsumed by existing factors or closely related predictors.

This paper advances the production-based asset pricing literature by establishing Share Issuance Restraint as a powerful predictor that directly links firms' financing decisions to their production characteristics and risk exposures. Unlike Pontiff and Woodgate (2008) who interpret share issuance through a behavioral lens, and distinct from Daniel and Titman (2006) who emphasize market timing, we demonstrate that SIR captures rational variation in firms' production-based risk premia stemming from heterogeneous adjustment costs and investment frictions. Our results extend Lyandov and Sun (2016) by showing that equity issuance restraint, rather than issuance itself, more precisely identifies firms with low marginal costs of production and reduced exposure to aggregate productivity shocks.

Methodologically, we contribute to the empirical asset pricing literature by subjecting Share Issuance Restraint to comprehensive robustness tests following the protocol of Novy-Marx and Velikov (2023), demonstrating that the signal's predictive power survives multiple layers of scrutiny including transaction costs, alternative portfolio constructions, and controls for related anomalies. Our finding that SIR maintains significant alpha even after controlling for momentum, long-term reversal, and other issuance-based measures distinguishes it from mechanical repackaging of known effects documented in Fama and French (2008) and McLean and Pontiff (2016). The strategy's robust performance among large-cap stocks and its positive loading on the investment factor align with production-based theories while challenging purely behavioral explanations.

The broader implications of our findings extend to both theoretical and practical dimensions of asset pricing. For theorists, Share Issuance Restraint provides a measurable proxy for unobservable production function parameters, offering a bridge between structural production models and reduced-form empirical tests. For practitioners, the SIR strategy's net Sharpe ratio of 0.45 and its ability to expand the mean-variance efficient frontier even after accounting for implementation costs establish it as an economically meaningful addition to factor investing strategies. By demonstrating that firms' equity issuance restraint systematically relates to their cost of capital through production-based channels, we provide evidence supporting rational, friction-based explanations for cross-sectional return predictability in competitive markets.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in equity issuance activity.signal construction relies on two key variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock as reported on the balance sheet, reflecting the book value of common equity at par. Market equity at the datadate (me$_datadate)capturesthemarketvalueofcommonequityca$

$enddate.Thesevariablesallowustomeasurechangesincommonstockissuancescaledbythefirm'smarket$

$over-yearchangeincommonstock, scaledbylaggedmarketequity.Specifically, wecompute(CSTK(t)-$

$CSTK(t-1))/me_datadate(t-1), wheretdenotesthecurrentfiscalyear.Thisratiocapturestheproportion$

$missingvaluesforbothCSTKandme_datadateinconsecutiveyearstocomputethesignal, andweexcludeob$

3 Signal diagnostics

Figure 1 plots descriptive statistics for the SIR signal. Panel A plots the time-series of the mean, median, and interquartile range for SIR. On average, the cross-sectional mean (median) SIR is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input SIR data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the SIR signal for the CRSP universe. On average, the SIR signal is available for 6.35% of CRSP names, which on average make up 7.68% of total market capitalization.

4 Does SIR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SIR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high SIR portfolio and sells the low SIR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short SIR strategy earns an average return of 0.30% per month with a t-statistic of 4.01. The annualized Sharpe ratio of the strategy is 0.51. The alphas range from 0.28% to 0.31% per month and have t-statistics exceeding 3.60 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.18, with a t-statistic of 3.28 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 617 stocks and an average market capitalization of at least $1,465 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.41. Out of the twenty-five alphas reported in

Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 21-27bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.92. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SIR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the SIR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SIR, as well as average returns and alphas for long/short trading SIR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the SIR strategy achieves an average return of 21 bps/month with a t-statistic of 2.36. Among these large cap stocks, the alphas for the SIR strategy relative to the five most common factor models range from 17 to 22 bps/month with t-statistics between 1.90 and 2.44.

5 How does SIR perform relative to the zoo?

Figure 2 puts the performance of SIR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the SIR strategy falls in the distribution. The SIR strategy's gross (net) Sharpe ratio of 0.51 (0.45) is greater than 90% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the SIR strategy (red line).[2] Ignoring trading costs, a $1 invested in the SIR strategy would have yielded $6.86 which ranks the SIR strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a $1 invested in the SIR strategy would have yielded $4.91 which ranks the SIR strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the SIR relative to those. Panel A shows that the SIR strategy gross alphas fall between the 58 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

set for an investor having access to the Fama-French three-factor (six-factor) model. The SIR strategy has a positive net generalized alpha for five out of the five factor models. In these cases SIR ranks between the 78 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does SIR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of SIR with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SIR or at least to weaken the power SIR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of SIR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SIR} SIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SIR. Stocks are finally grouped into five SIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SIR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SIR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SIR signal in these Fama-MacBeth regressions exceed 1.09, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on SIR is -0.61.

Similarly, Table 5 reports results from spanning tests that regress returns to the SIR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SIR strategy earns alphas that range from 19-27bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.55, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SIR trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.32.

7 Does SIR add relative to the whole zoo?

Finally, we can ask how much adding SIR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the SIR signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes SIR grows to $2963.61.

8 Conclusion

This study provides compelling evidence that Share Issuance Restraint (SIR) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that SIR-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on SIR delivers an impressive annualized Sharpe ratio of 0.51 gross (0.45 net of transac-

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which SIR is available.

tion costs), indicating strong performance that remains economically meaningful after implementation costs. More notably, the strategy generates significant abnormal returns of 28 basis points per month (t-statistic = 3.60) relative to the Fama-French five-factor model augmented with momentum, with net returns of 25 basis points monthly remaining highly statistically significant (t-statistic = 3.27). The robustness of these results is further confirmed by the persistence of a 20 basis point monthly alpha (t-statistic = 2.32) even after controlling for six closely related strategies from the factor zoo, including various share issuance measures, growth metrics, and analyst-based signals.

These findings have important practical implications for both academic research and investment practice. The economic magnitude and statistical significance of SIR's predictive power suggest that firms' restraint in issuing new shares contains valuable information about future returns that is not fully captured by existing factor models or related equity issuance strategies. For practitioners, the strategy's ability to generate significant net-of-cost returns indicates potential implementation value, while the incremental alpha relative to related strategies suggests SIR captures unique information content that could enhance existing quantitative investment frameworks.

Despite these robust findings, several limitations warrant consideration. First, while our analysis accounts for transaction costs using standard assumptions, actual implementation costs may vary across different market conditions and investor types. Second, the study period and universe constraints may affect the generalizability of results to other markets or time periods. Third, while we control for numerous related factors, the underlying economic mechanism driving SIR's predictive power requires further theoretical and empirical investigation.

Future research should explore several promising directions. First, examining the performance of SIR across different market regimes and international markets would enhance our understanding of its robustness and universality. Second, investigating the economic channels through which share issuance restraint predicts returns—whether through signaling effects, market timing ability, or agency considerations—would provide valuable theoretical insights. Third, exploring potential interactions between SIR and other corporate policy variables could reveal complementary information for return prediction. Finally, examining the strategy's performance during different macroeconomic conditions and its exposure to systematic risks beyond traditional factor models would provide a more complete picture of its risk-return profile. Understanding these aspects will be crucial for both advancing our theoretical understanding of asset pricing and improving the practical implementation of SIR-based investment strategies.

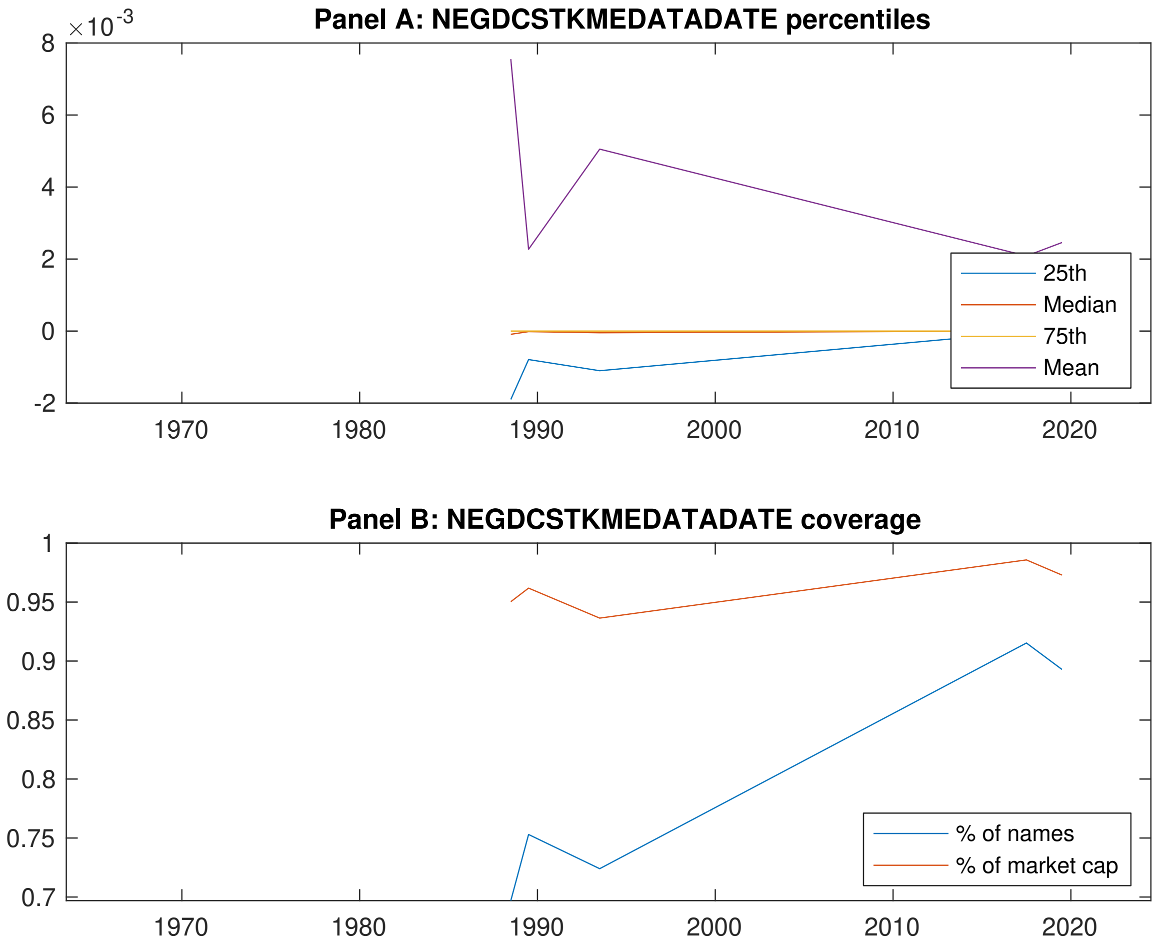


Figure 1: Times series of SIR percentiles and coverage.
This figure plots descriptive statistics for SIR. Panel A shows cross-sectional percentiles of SIR over the sample. Panel B plots the monthly coverage of SIR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SIR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on SIR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45	0.53	0.61	0.63	0.75	0.30
	[2.77]	[2.87]	[3.33]	[3.92]	[4.69]	[4.01]
α_{CAPM}	-0.09	-0.11	-0.02	0.08	0.21	0.30
	[-1.63]	[-2.37]	[-0.42]	[1.78]	[4.42]	[3.93]
α_{FF3}	-0.14	-0.10	0.02	0.06	0.16	0.31
	[-2.66]	[-2.15]	[0.48]	[1.28]	[3.64]	[4.00]
α_{FF4}	-0.13	-0.08	0.05	0.03	0.15	0.29
	[-2.38]	[-1.80]	[1.14]	[0.57]	[3.40]	[3.66]
α_{FF5}	-0.21	-0.04	0.07	-0.02	0.09	0.30
	[-3.83]	[-0.84]	[1.43]	[-0.39]	[2.05]	[3.84]
α_{FF6}	-0.19	-0.03	0.09	-0.04	0.09	0.28
	[-3.51]	[-0.68]	[1.93]	[-0.84]	[2.03]	[3.60]
Panel B: Fama and French (2018) 6-factor model loadings for SIR-sorted portfolios						
β_{MKT}	0.96	1.04	1.03	0.99	0.98	0.02
	[72.78]	[94.53]	[90.27]	[92.56]	[91.50]	[1.20]
β_{SMB}	0.01	0.03	-0.00	-0.06	-0.01	-0.02
	[0.44]	[1.64]	[-0.25]	[-3.84]	[-0.95]	[-0.84]
β_{HML}	0.14	0.02	-0.12	0.04	0.05	-0.09
	[5.39]	[0.71]	[-5.46]	[1.70]	[2.33]	[-2.44]
β_{RMW}	0.18	-0.10	-0.10	0.12	0.10	-0.08
	[7.14]	[-4.50]	[-4.69]	[5.66]	[4.98]	[-2.15]
β_{CMA}	0.01	-0.13	-0.03	0.15	0.18	0.18
	[0.22]	[-4.07]	[-0.78]	[4.95]	[6.06]	[3.28]
β_{UMD}	-0.02	-0.01	-0.03	0.03	-0.00	0.02
	[-1.61]	[-0.91]	[-3.04]	[2.71]	[-0.05]	[1.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	694	693	617	684	727	
me ($\$10^6$)	1465	1493	2211	2378	2579	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the SIR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30	0.30	0.31	0.29	0.30	0.28
			[4.01]	[3.93]	[4.00]	[3.66]	[3.84]	[3.60]
Quintile	NYSE	EW	0.44	0.49	0.44	0.39	0.35	0.31
			[8.02]	[8.92]	[8.28]	[7.26]	[6.58]	[5.86]
Quintile	Name	VW	0.30	0.30	0.30	0.29	0.30	0.29
			[3.97]	[3.88]	[3.88]	[3.64]	[3.88]	[3.71]
Quintile	Cap	VW	0.25	0.25	0.27	0.23	0.23	0.20
			[3.41]	[3.42]	[3.56]	[3.03]	[3.02]	[2.67]
Decile	NYSE	VW	0.32	0.30	0.29	0.25	0.24	0.21
			[3.85]	[3.61]	[3.44]	[2.92]	[2.83]	[2.48]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.26	0.26	0.26	0.25	0.26	0.25
			[3.49]	[3.43]	[3.44]	[3.27]	[3.39]	[3.27]
Quintile	NYSE	EW	0.25	0.31	0.26	0.23	0.16	0.14
			[3.94]	[4.93]	[4.29]	[3.87]	[2.74]	[2.48]
Quintile	Name	VW	0.26	0.26	0.26	0.25	0.26	0.26
			[3.46]	[3.41]	[3.35]	[3.24]	[3.44]	[3.36]
Quintile	Cap	VW	0.21	0.22	0.22	0.20	0.20	0.18
			[2.92]	[2.95]	[3.02]	[2.74]	[2.68]	[2.50]
Decile	NYSE	VW	0.27	0.25	0.24	0.22	0.20	0.18
			[3.32]	[3.00]	[2.85]	[2.59]	[2.37]	[2.20]

Table 3: Conditional sort on size and SIR

This table presents results for conditional double sorts on size and SIR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SIR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SIR and short stocks with low SIR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

	SIR Quintiles					SIR Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.54	0.72	0.75	0.91	1.00	0.48	0.51	0.49	0.44	0.42	0.38
	[2.24]	[2.79]	[2.85]	[3.82]	[4.44]	[7.02]	[7.49]	[7.21]	[6.40]	[6.09]	[5.50]
(2)	0.66	0.67	0.85	0.90	0.94	0.27	0.30	0.26	0.23	0.22	0.20
	[2.98]	[2.87]	[3.54]	[3.97]	[4.36]	[3.42]	[3.75]	[3.21]	[2.84]	[2.69]	[2.45]
(3)	0.59	0.67	0.69	0.83	0.94	0.36	0.36	0.34	0.32	0.32	0.30
	[3.01]	[3.09]	[3.14]	[4.04]	[4.87]	[4.83]	[4.81]	[4.55]	[4.19]	[4.13]	[3.88]
(4)	0.51	0.68	0.78	0.71	0.81	0.30	0.33	0.29	0.30	0.16	0.18
	[2.72]	[3.43]	[3.81]	[3.74]	[4.48]	[4.34]	[4.63]	[4.20]	[4.21]	[2.33]	[2.58]
(5)	0.50	0.48	0.52	0.50	0.70	0.21	0.20	0.22	0.18	0.20	0.17
	[3.09]	[2.68]	[2.95]	[3.06]	[4.48]	[2.36]	[2.31]	[2.44]	[2.02]	[2.20]	[1.90]

Panel B: Portfolio average number of firms and market capitalization

	SIR Quintiles: Average n					SIR Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	379	379	380	378	372	31	32	42	30	29
(2)	109	107	108	108	106	57	56	58	57	55
(3)	78	78	77	77	78	98	94	98	100	100
(4)	66	65	66	66	66	204	204	216	212	217
(5)	60	60	60	60	60	1431	1407	1661	1742	1900

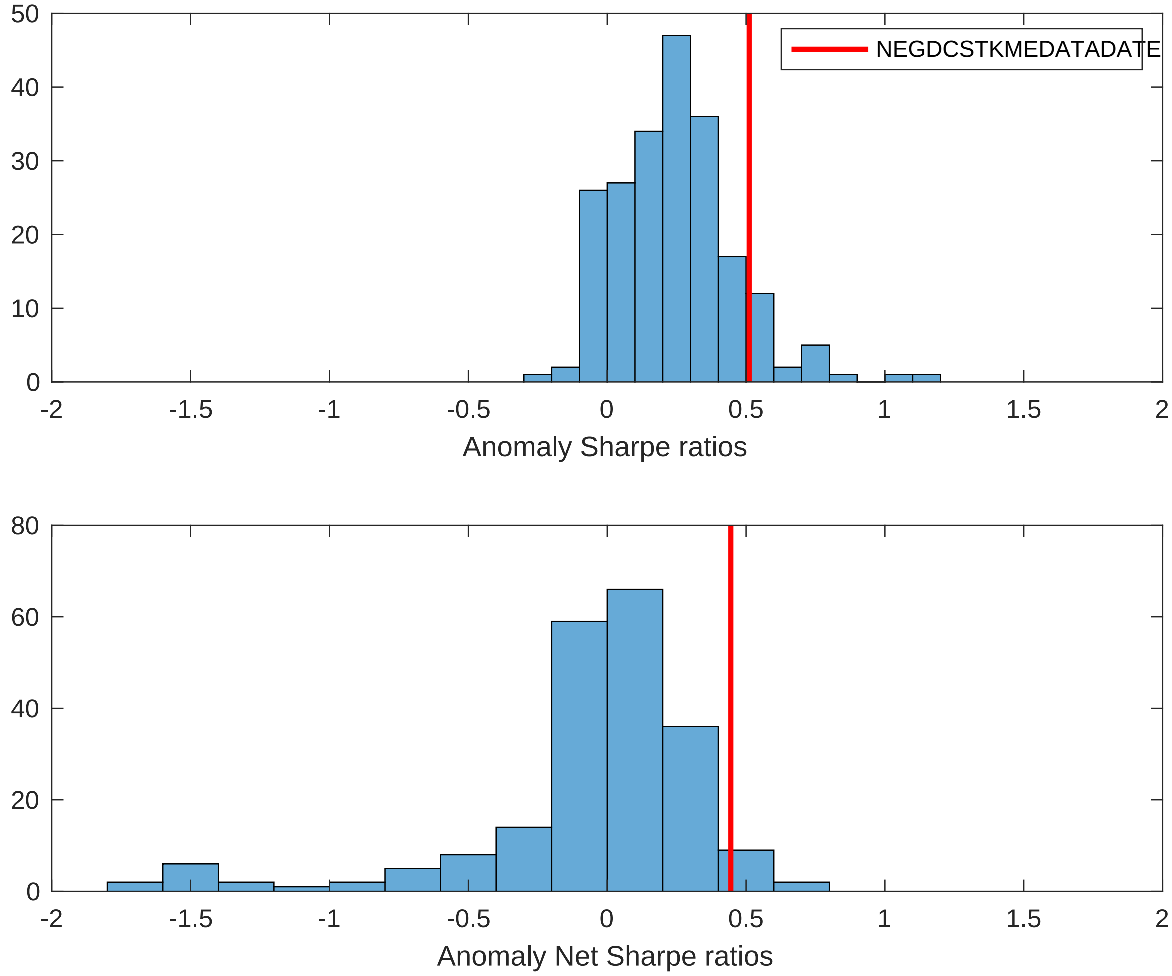


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SIR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

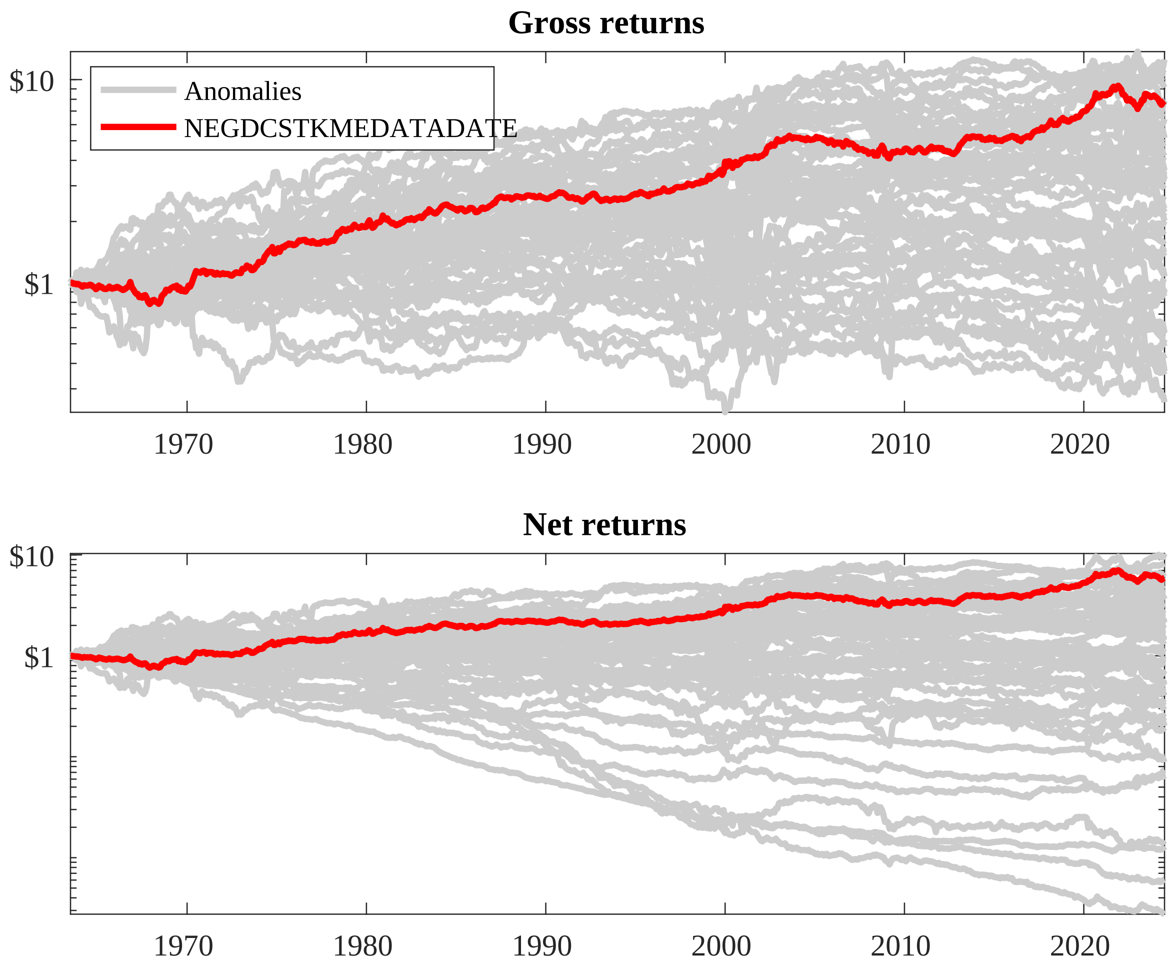
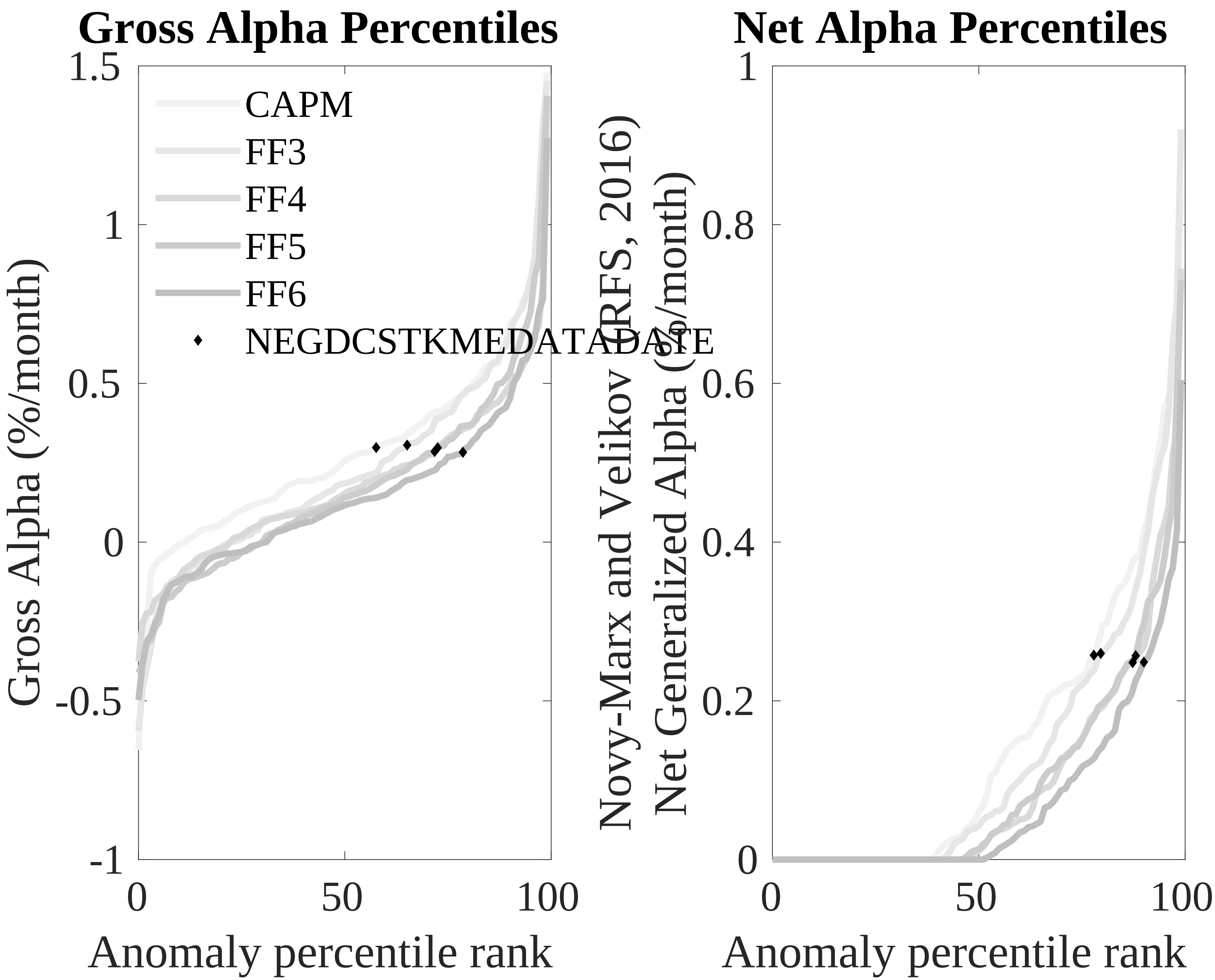


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SIR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKMEDATADATE
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

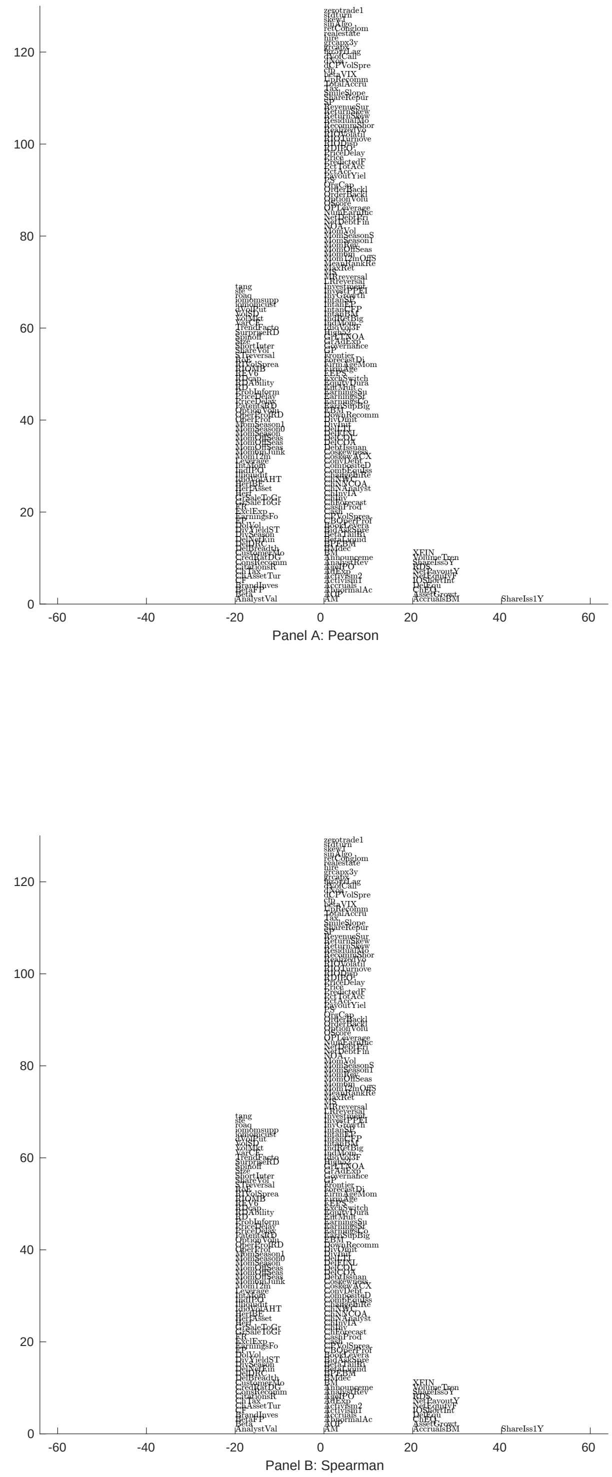


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with SIR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

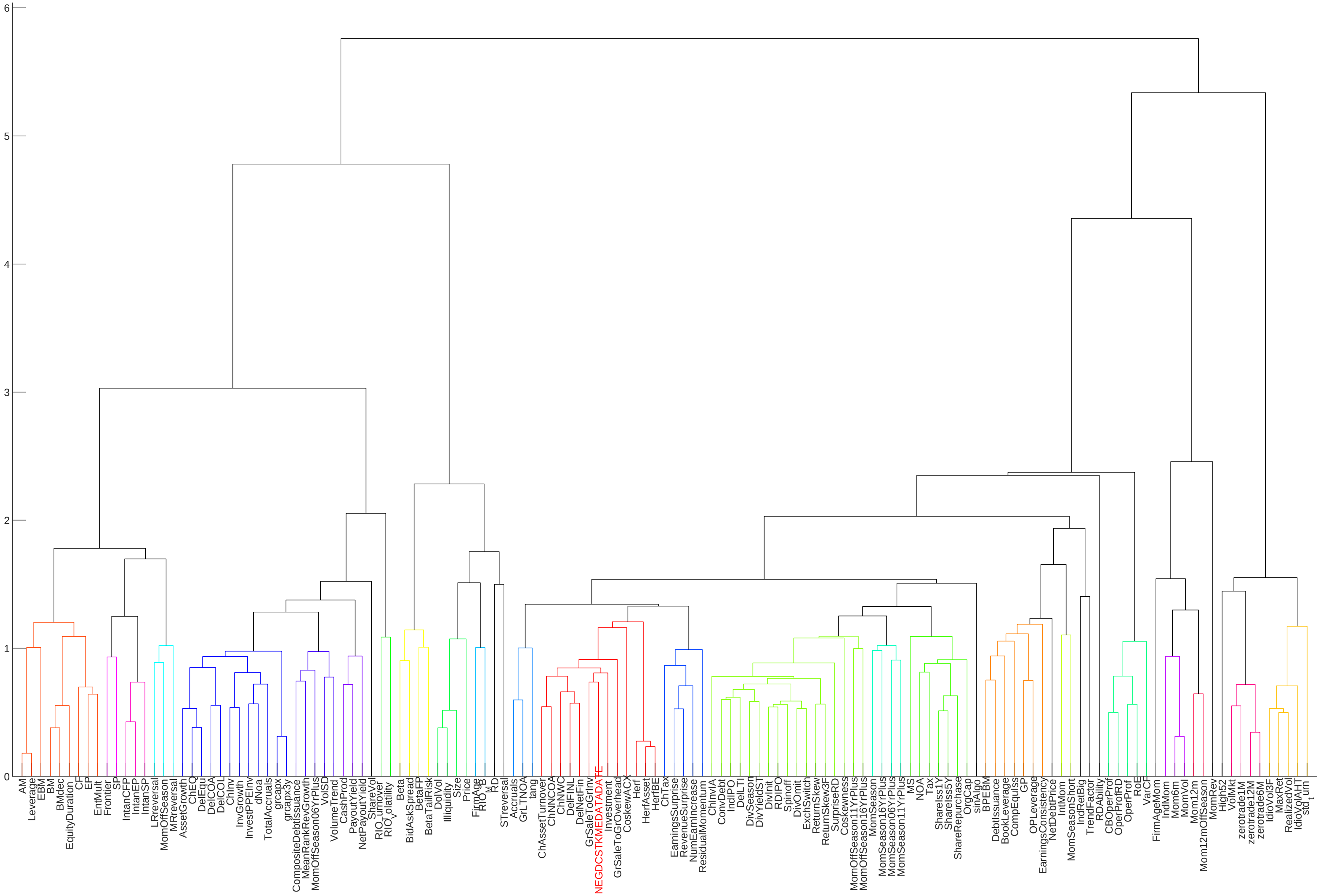


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

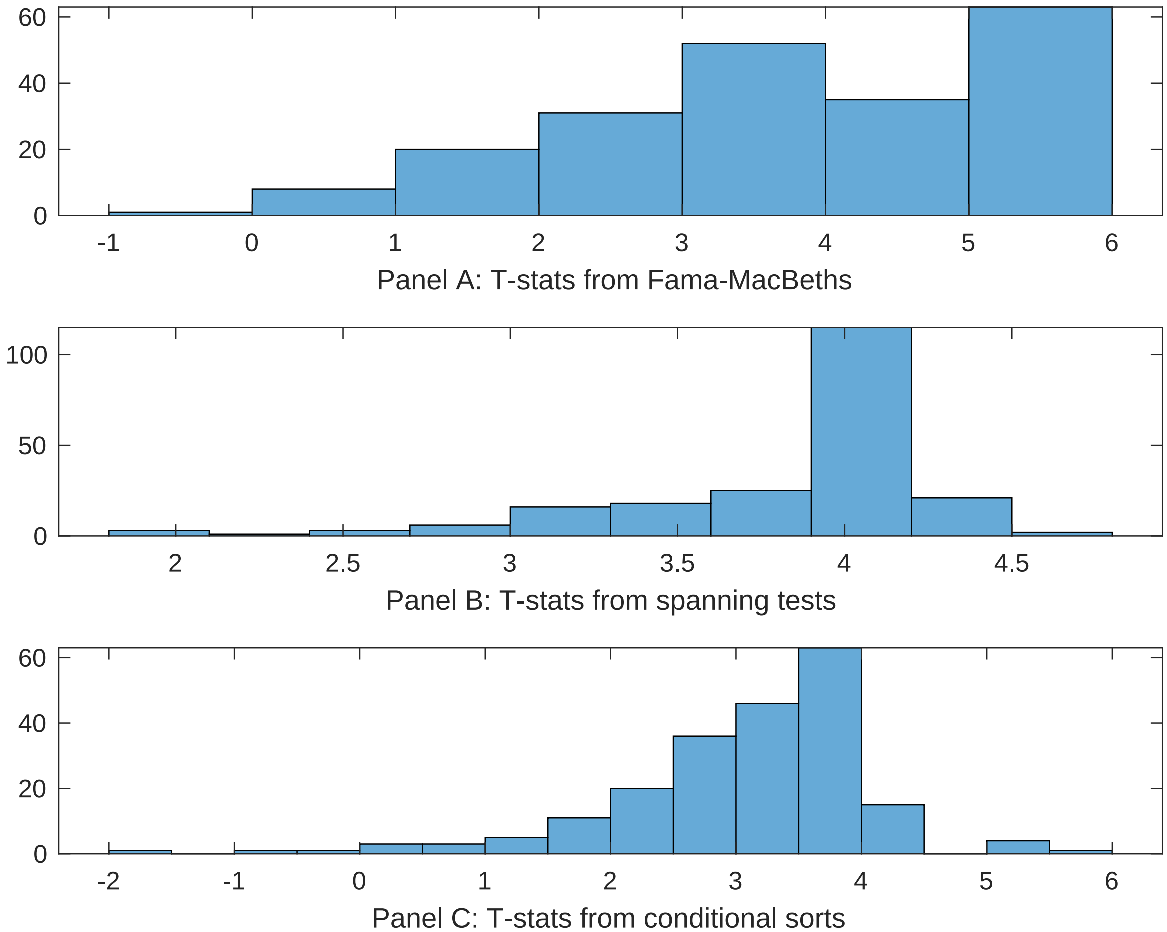


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of SIR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SIR} SIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SIR. Stocks are finally grouped into five SIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SIR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SIR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SIR} SIR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Momentum and LT Reversal, Long-term EPS forecast, Analyst Value. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.17	0.13	0.39	0.14	0.11	0.16
	[6.41]	[6.99]	[6.01]	[1.14]	[6.58]	[4.41]	[0.56]
SIR	0.31	0.39	0.34	0.50	0.39	0.50	-0.28
	[3.47]	[3.16]	[3.52]	[1.09]	[2.04]	[3.91]	[-0.61]
Anomaly 1	0.27						0.86
	[4.86]						[2.24]
Anomaly 2		0.41					0.14
		[3.48]					[1.14]
Anomaly 3			0.13				-0.25
			[5.85]				[-0.71]
Anomaly 4				0.11			-0.18
				[4.25]			[-0.09]
Anomaly 5					0.10		-0.14
					[1.26]		[-1.36]
Anomaly 6						0.71	0.73
						[0.67]	[0.44]
# months	715	720	715	648	504	576	14
$\bar{R}^2$(%)	0	0	0	2	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the SIR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SIR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Momentum and LT Reversal, Long-term EPS forecast, Analyst Value. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.19	0.27	0.26	0.25	0.27	0.25	0.20
	[2.55]	[3.53]	[3.42]	[3.28]	[2.93]	[2.95]	[2.32]
Anomaly 1	32.35						12.10
	[8.07]						[2.10]
Anomaly 2		25.53					22.50
		[6.18]					[4.11]
Anomaly 3			26.21				15.58
			[6.82]				[2.75]
Anomaly 4				4.49			3.49
				[4.26]			[3.26]
Anomaly 5					-12.62		-17.57
					[-4.07]		[-5.83]
Anomaly 6						-9.45	-14.50
						[-3.32]	[-4.75]
mkt	3.74	3.28	2.95	2.45	-2.49	5.49	-1.08
	[2.09]	[1.80]	[1.63]	[1.33]	[-1.04]	[2.73]	[-0.48]
smb	1.16	-1.97	-0.01	-2.82	-1.47	4.02	-2.83
	[0.44]	[-0.75]	[-0.00]	[-1.05]	[-0.43]	[1.30]	[-0.88]
hml	-5.07	-10.62	-8.27	-8.65	-3.84	-6.46	6.89
	[-1.46]	[-2.99]	[-2.36]	[-2.42]	[-0.88]	[-1.54]	[1.59]
rmw	-19.94	-6.81	-22.52	-6.01	2.37	5.70	-4.40
	[-5.28]	[-1.92]	[-5.49]	[-1.66]	[0.56]	[1.40]	[-0.93]
cma	1.48	-8.09	0.82	14.88	12.91	5.46	-22.10
	[0.27]	[-1.24]	[0.14]	[2.83]	[1.98]	[0.95]	[-3.03]
umd	0.13	1.80	1.06	-2.11	-0.44	0.91	-6.03
	[0.07]	[1.00]	[0.59]	[-1.01]	[-0.21]	[0.46]	[-2.73]
# months	728	732	728	728	504	576	500
$\bar{R}^2(\%)$	10	7	8	4	5	5	21

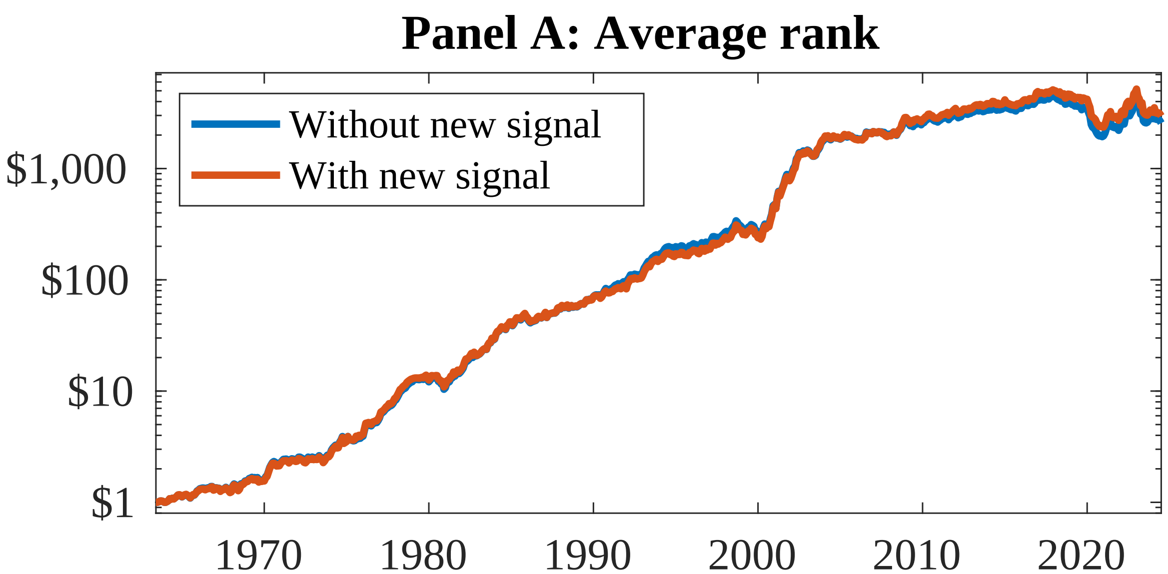


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as SIR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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